LABEL THE DIAGRAM

Answers

LIVING LIMITS · PART 1.2

1 Tolerance range (9–25°C) · **2** Optimum range (13–21°C) · **3** Lower limit (9°C) · **4** Upper limit (25°C) · **5** Zone of physiological stress · **6** Zone of intolerance

FISH TO PLANET · PART 1.3

1 Organism · **2** Population · **3** Biological community · **4** Ecosystem · **5** Biome · **6** Biosphere

MAKERS & TAKERS · PART 2.1

1 Sun (energy source) · **2** Producer · **3** Herbivore · **4** Omnivore · **5** Carnivore · **6** Arrow = direction of energy flow · **7** Decomposer

THE 10% RULE · PART 2.2

1 Third-level consumers (0.1%) · **2** Secondary consumers (1%) · **3** Primary consumers (10%) · **4** Primary producers (100%) · **5** Energy lost as heat · **6** Available energy decreases

ROUND & ROUND · PART 3.1

1 Solar energy · **2** Evaporation · **3** Transpiration · **4** Condensation · **5** Precipitation · **6** Runoff · **7** Percolation in soil · **8** Groundwater (aquifer) · **9** Lake · **10** Ocean

CARBON LOOP · PART 3.2

1 CO_2 in atmosphere · **2** Photosynthesis · **3** Respiration · **4** Combustion · **5** Decay · **6** Fossil fuels (coal, oil, gas) · **7** Dead organisms and animal waste · **8** Diffusion (air ⇄ ocean) · **9** Sedimentation (shells → limestone) · **10** Ocean · **11** Land plants

FIX THE AIR · PART 3.3

1 N_2 in atmosphere · **2** Nitrogen-fixing bacteria (root nodules) · **3** Ammonium (NH_4^+) · **4** Nitrite (NO_2^-) · nitrifying bacteria · **5** Nitrate (NO_3^-) · **6** Plants (assimilation) · **7** Denitrification · **8** Decomposers · **9** Lightning · **10** Consumers

ROCK TO ROOT · PART 3.4

1 Mineable rock · **2** Weathering · **3** Phosphate in soil · **4** Plants (producers) · **5** Decomposers · **6** Fertilizer · **7** Phosphate in solution · **8** Organisms (in water) · **9** Sedimentation · **10** Uplift (long-term)

TEST IT · PART 1.2

1 State the problem · **2** Gather information · **3** Form a hypothesis · **4** Test the hypothesis · **5** Analyze data · **6** Draw conclusions · **7** Hypothesis is not supported · **8** Hypothesis is supported